

DuPont Tedlar AI Lead Generation System

1. Solution Architecture

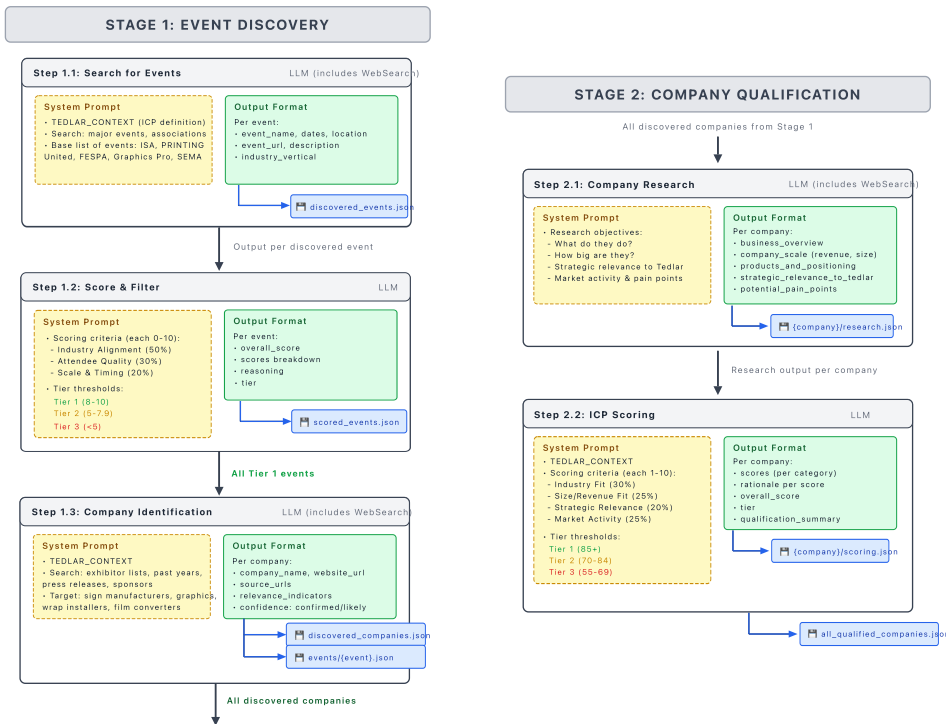
A **four-stage AI GTM workflow** that progressively identifies, scores and filters leads:

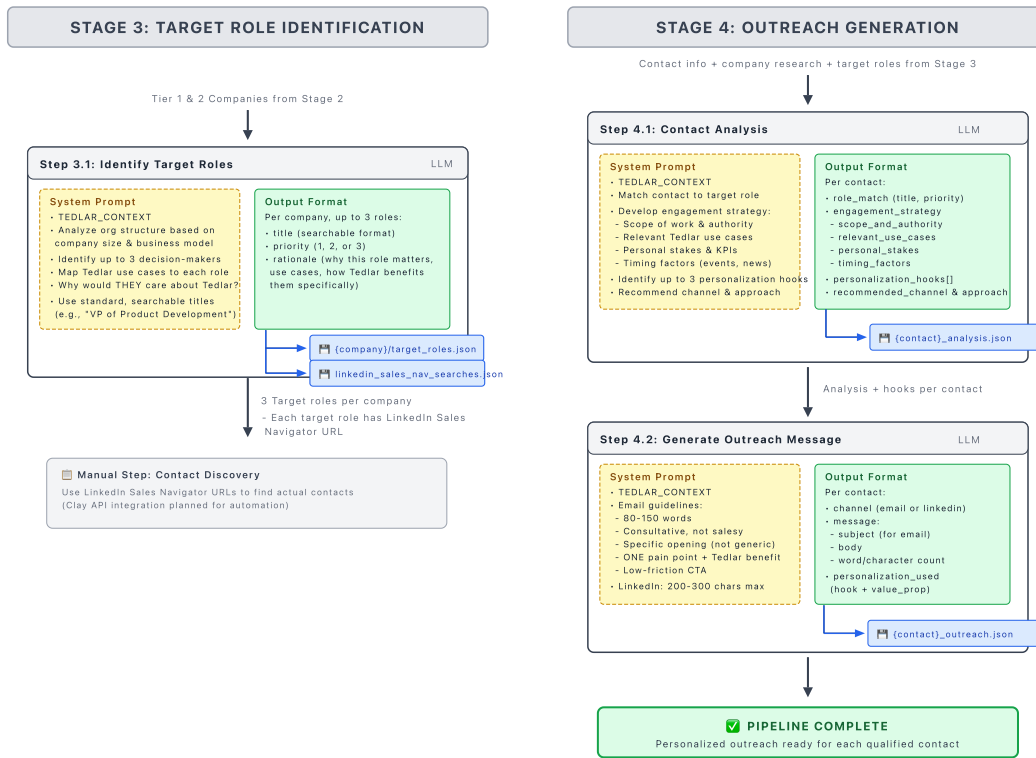


Key Design Decisions

- **Event-driven orchestration:** Completing in-progress work is prioritized over starting new. When Stage 2 finishes, Stage 3 automatically triggers.
- **Tiered qualification:** Scoring with explicit thresholds ensures resources focus on high-potential leads. Low-scoring companies filtered early.
- **File-based state:** Each stage produces JSON artifacts. Enables resume from interruption without database dependencies.

2. Data Processing Pipeline





3. AI Workflow Orchestration

The pipeline is managed by an `main.py`, the main orchestrator that relies on event triggers with:

- Priority order:** Later stages processed first—companies closer to completion are finished before starting new ones.
- Automatic triggers:** Callbacks connect stages (when Stage 2 completes, Stage 3 starts automatically for qualified companies).
- Rate limiting:** Shared limiter tracks API tokens (30k/minute) across all stages.
- Resume capability:** File existence checks allow safe restart after interruption.

4. Implementation Results

4.1 Data Outputs

Each stage produces JSON artifacts stored in `data/`:

```
data/
├── events/
│   ├── discovered_events.json    # 22 raw events from Stage 1.1
│   ├── scored_events.json       # Tiered events from Stage 1.2
│   ├── discovered_companies.json # 37 companies from Stage 1.3
│   └── companies/
│       └── {EventName}.json      # Companies per event
├── companies/
│   ├── tier_1_high_priority.json  # Summary of top companies
│   ├── all_qualified_companies.json
│   └── {CompanyName}/
│       └── research.json         # Stage 2.1 deep research
```

```
├── scoring.json          # Stage 2.2 ICP scores
├── target_roles.json     # Stage 3 decision-makers
├── linkedin_searches.json # Sales Nav search URLs
└── contacts/
    ├── role_identification_results.json
    ├── linkedin_sales_nav_searches.json
    └── {CompanyName}/
        ├── {ContactName}_analysis.json # Stage 4.1 hooks/angles
        └── {ContactName}_outreach.json  # Stage 4.2 email draft
```

5. Limitations & Future Work

Current Limitations:

- Contact discovery requires manual LinkedIn searches (Clay API integration planned but not implemented yet)
- Rate limits require ~60s min. delays between API calls

Potential Enhancements:

- Integrate Clay API for automated contact enrichment
- Add email verification step before outreach
- CRM integration for direct lead export